LEPTONS AND KOIDE: THREE ICOSAHEDRAL MODES OF THE JOBSON CELL LATTICE

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supersedes doc_muon.txt, doc_tau.txt
branches from doc_alpha, doc_jobson_cell
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ABSTRACT

The Standard Model contains exactly three lepton generations with no explanation for why there are three, nor for their mass ratios. Upon discovering the muon in 1936, Rabi asked "who ordered that?" -- and the question has never been answered. The tau was discovered in 1975, and the pattern remains unexplained: the SM accepts the three generations as input, it does not derive them.
This paper derives all three from geometry alone.

All three lepton generations are derived from the three geometric element types of the icosahedron (V=12 vertices, E=30 edges, F=20 faces) to sub-0.01% precision with zero free parameters.

```
ELECTRON (E+,  dim=2, C3=-1): VERTEX mode
  Scatters at icosahedral vertices. Deflection = 72 deg (C5). eff_e = phi.
  m_e  = 0.510999 MeV  (PDG: 0.510999 MeV,  +0.000065%)

MUON (G32, dim=4, C3=+1): EDGE mode
  Traverses icosahedral edges. Deflection = 72 deg (C5). eff_mu = (9-sqrt5)/8.
  m_mu = 105.655 MeV   (PDG: 105.658 MeV,    -0.003%)

TAU (I52, dim=6, C3=0): FACE mode
  Hamiltonian circuit of all 20 face-center NEXUSES (at inradius = 0.756 x L_J
  on the outer surface of the cell). Between nexuses the tau travels as a
  STRAIGHT-LINE CHORD through the cell interior: at each crossing the chord
  dips to r = 0.706 x L_J, which is 0.103 x L_J INSIDE the edge midpoint
  (r_mid = 0.809 x L_J) -- no edge-clearing needed; the path passes beneath
  the edge geometrically. The face-center nexuses face inward; the tau
  arrives at each one from the interior and scatters off the convergent
  gluon maximum (all 3 gluons of that face reach amplitude peak simultaneously
  [GH0b]), turning 72 deg (C5, same as muon) [GH2]. Alpha=0 (no edge contact)
  is a geometric consequence, not an assertion.
  The 138.19-deg dihedral (JC5) is the fold angle of the cell surface at each
  edge the chord crosses -- not the tau's turning angle.
```

The Koide formula (m_e+m_mu+m_tau)/(sqrt(m_e)+sqrt(m_mu)+sqrt(m_tau))^2 = 2/3 holds to 9.2 ppm for measured masses. With our derived m_e and m_mu, it predicts m_tau to 0.004%. The Koide 2/3 has a proven geometric origin: it is the field/field-strength ratio of the icosahedral cell (dim(T_1g+T_2g)/dim(T_1g x T_2g) = 6/9 = 2/3 exactly). The algebraic connection to the exact Born balance values remains the final open step.

NEW GROUND: the Koide formula's exact value of 2/3 has been an unexplained empirical coincidence since its discovery (Koide, 1981) despite decades of

attempted explanations (radiative corrections, family symmetry textures, preon models). This paper is the first to obtain 2/3 as an exact group-theoretic dimension ratio of a specific finite symmetry group, rather than as a numerical fit or an assumed mass-texture ansatz.

PHYSICAL ROLES IN THE JOBSON CELL MEDIUM:

```
The Jobson cell is the icosahedral unit cell of the torsion medium. Its
structure is made of torsion field modes (A_g, T_1g, T_2g, 2G, H_g).
The leptons are RESONANT MODES that propagate THROUGH the cell structure
-- they are shaped by the cell geometry but are not part of the cell itself.
Analogy: the cell is the crystal lattice; leptons are the phonon/acoustic modes.

CELL STRUCTURE (what the cell IS made of -- see doc_jobson_cell Section 7):
  Higgs (A_g):   isotropic bulk mode (K channel), jamming = SSB; also the
                 v_p/GW170817 longitudinal branch [doc_torsion Sec 3.1]
  T_1g:          transverse vector field (W/Z live here) -- NOT compression
  T_2g:          ELASTIC FACE MATERIAL -- color-neutral shear field, creates
                 the triangular face panels. Rs^2 = T_2g shear amplitude squared
                 at the Maxwell critical point. Rs^2 appears in lepton mass
                 corrections because lepton modes interact with the T_2g surface.
  2G (gluons):   color-active field ON the T_2g faces (8 modes, C3=+1)

LEPTON MODES (what propagates through the cell structure):
  ELECTRON (E+): scatters at the 12 vertices where T_2g face panels converge
  MUON (G32):    zig-zag along edges -- probes the T_1g/T_2g face boundary tension
  TAU  (I52):    Hamiltonian circuit of 20 face-center NEXUSES at inradius
                 (0.756 x L_J). Between nexuses the tau travels as a CHORD
                 through the cell interior, dipping to r = 0.706 x L_J at
                 mid-hop -- 0.103 x L_J inside the edge midpoint (0.809 x L_J).
                 The face-center nexuses face inward; the tau arrives from
                 the interior, scatters off the convergent gluon maximum
                 (all 3 gluons of that face peak simultaneously [GH0b]),
                 and turns 72 deg (C5, same as muon) [GH2].
                 Alpha=0 (no edge contact) is geometric: the chord passes
                 0.103 x L_J beneath the edge -- no surface-clearing needed.
                 Three events per hop:
                   face-center nexus: tau hits inward-facing gluon maximum
                                      -> 72-deg C5 turn (eff_tau coupling)
                   chord segment:     straight line through empty interior
                   edge fold:         surface dihedral 138.19 deg (JC5),
                                      geometry of the shell -- not tau angle
                 m_tau = (phi^3/sqrt5)*m_p = 20 face-center gluon nexuses.
```

The original pentagonal bipyramid model (path: top->eq->bottom->eq->...) was accidentally a SUPERPOSITION of the muon and tau modes: its apex deflections (72 deg) encode the muon, and its equatorial deflections (116.57 deg) encode the tau. The pure paths are separate: muon has C5 vertex nexuses (72-deg turns); tau has C5 face-center nexuses (72-deg turns, gluon-maximum coupling, GH2).

Companion script: analysis/quantum/lepton_mass.py (18/18 PASS)

1. IDENTIFICATION FROM 2I DOUBLE GROUP

1.1 The 2I irreducible representations

The icosahedral double group 2I has 9 irreps:

```
A(1), E+(2), E-(2), T1(3), T2(3), G32(4), G(4), H(5), I52(6)
```

The lepton generations are assigned by spinor rank (half-integer j):

```
E+  (dim=2, j=1/2): electron
```

```
G32 (dim=4, j=3/2): muon
I52 (dim=6, j=5/2): tau
```

The dimensions form the arithmetic sequence 2, 4, 6 -- all half-integer spinors in 2I, exhausting the sequence at j=5/2 (I52 is the highest spinor irrep in 2I).

1.2 Second-order Born corrections (why no alpha^2 for muon or tau)

From 2I CG decompositions [ih_double_cg.py DC2, DC6, DC7 -- actual computed products, not just multiplicities]:

```
E+ x E+   = A(1) + T1(3)                       -> coefficient = dim(T1)/4 = 3/4
G32 x G32 = A(1)+T1(3)+T2(3)+G(4)+H(5)         -> T1=T2 (1x each) cancel
I52 x I52 = A(1)+2*T1(3)+2*T2(3)+2*G(4)+3*H(5) -> T1=T2 (2x each) cancel
```

T1 (pressure, same as T_1g cell mode) and T2 (shear, same as T_2g cell mode) appear in EQUAL multiplicity for G32 and I52 (so their contributions cancel), but T2 is ABSENT for E+ (only A+T1 appear, no T2 term at all) -- this asymmetry is why the electron alone gets a nonzero alpha^2 correction, with coefficient = dim(T1)/dim(E+xE+) = 3/(1+3) = 3/4 directly (no separate "NET" subtraction step is needed or used).

The muon's -0.003% and the tau's +0.004% are each individually-computed residuals from the current (approximate) Born-balance inputs; whether they "close simultaneously when the exact Born balance is applied to all paths" is the still-open calculation listed in Section 9 (exact muon Born balance, eff_mu=0.8563 numerical) -- not yet shown, despite the more confident phrasing this sentence previously had.

2. THE ICOSAHEDRAL ELEMENT MAPPING

2.1 Three elements, three leptons

The icosahedron has three classes of geometric element:

```
V = 12 vertices: each vertex is a junction of 5 triangular faces.
                 C5 rotation axes pass through opposite vertex pairs.
E = 30 edges:    each edge is a boundary between 2 adjacent faces.
                 C5 axes pass through midpoints of opposite edge pairs.
F = 20 faces:    each face is one triangular region.
                 C3 rotation axes pass through opposite face centers.
```

Euler formula: V - E + F = 12 - 30 + 20 = 2

Maxwell criterion: 3V - E = 3(12) - 30 = 6 (marginally rigid lattice -- critical)

The three 2I spinors map to these three element types:

```
E+  -> VERTEX  (dim=2, C3 char=-1): anti-phase under C3 -- vertex is where
       5 faces meet, with a -1 phase from destructive interference of the 3-fold
       face contributions. Deflection = 72 deg = C5 rotation angle.

G32 -> EDGE    (dim=4, C3 char=+1): in-phase under C3 -- edge borders 2 faces,
       positive C3 coupling. Still deflects by 72 deg (C5 type, same as vertex)
       but the path structure is along EDGES not between vertices.

I52 -> FACE    (dim=6, C3 char=0):  zero C3 -- face is one colored region.
       Nexus at the face CENTER (geometric centroid, outer surface at inradius):
```

```
all three gluons of the face reach amplitude maximum there [GH0b].
Tau turns 72 deg at each face-center nexus (GH2, C5 -- same as muon).
The 138.19-deg dihedral (JC5) is the fold angle of the icosahedron
surface at each edge crossing between nexuses, not the tau turning angle.
The tau never contacts the edge itself (alpha=0).
```

2.2 Deflection angles

```
Vertex mode (electron, muon*):
  cos(theta) = 1/(2*phi) = cos(72 deg)    positive = forward-biased
  The C5 rotation angle. Shallow deflection; net forward propagation.
  *The muon's edge path also deflects at 72 deg (see Section 4).

Face mode (tau):
  cos(theta) = -1/sqrt(5) = cos(116.57 deg)   negative = backward-biased
  BIPYRAMID EQUATORIAL deflection (used for eff_tau Born balance parameter).
  NOTE: the actual ICOSAHEDRAL FACE-FACE DIHEDRAL = arccos(-sqrt(5)/3) = 138.19 deg
  [JC5]. The -1/sqrt5 is the dodecahedron dihedral / equatorial bipyramid deflection,
  not the icosahedral face dihedral. The tau mass uses Koide, not this angle.
  Steep deflection; net inward (backward) component -> heavier lepton.

The DIFFERENCE IN DISPLACEMENT per bounce:
  Muon: cos = +0.309 -> deflection 72 deg, forward component survives. The wave
        "slides along" the edge direction with a moderate turn at each vertex.
  Tau:  cos = -0.447 -> deflection 116.57 deg, backward component dominates.
        The wave "bounces back inward" at each face boundary. More compact orbit.
  This is a STEEPER deflection than the muon's, despite the tau having the same
  icosahedral host geometry -- see Section 5.2 for how this translates to mass
  (backward bias actually LOWERS the Born coupling amplitude; the tau's larger
  bounce count N=20 vs the muon's N=5 is what makes up the difference).
```

2.3 The bipyramid bridge (why the original model worked)

The regular pentagonal bipyramid (Johnson solid J13, all equal edges, h_t/r_e = 1/phi) was the first path model tested for the muon. It has TWO types of bounce:

```
Path: top_apex -> eq[k] -> bottom_apex -> eq[k+2] -> top_apex -> ...
At top/bottom apex:  cos = 1/(2*phi) = 72 deg  [C5 turn = MUON deflection]
At equatorial ring:  cos = -1/sqrt(5) = 116.57 deg  [bipyramid equatorial deflection;
                      NOT icosahedral face dihedral (that is 138.19 deg, JC5)]
```

The bipyramid path was accidentally a SUPERPOSITION of both modes:

```
- Its apex bounces encode the MUON (C5 edge turns)
- Its equatorial bounces encode the TAU (dihedral face crossings)
```

The Born balance for this mixed path: eff_mu = (1 + ratio)/2 where

```
ratio = |cos_apex| / |cos_eq| = (1/(2*phi)) / (1/sqrt5) = sqrt5/(2*phi)
eff_bipyramid = (9-sqrt5)/8 = 0.845
```

This eff gave -0.003% for the muon mass. It is the Born balance BETWEEN the two modes mixed in the bipyramid path, not the pure muon balance. The -0.003% residual is partly from using this approximate mixed-mode eff instead of the exact single-mode balance.

GEOMETRIC CORRECTION (verified in lepton_mass.py LM4b):

```
The actual icosahedral zig-zag path (following real icosahedral edges) has
ALL 5 deflections = cos(72 deg) = 1/(2*phi). The equatorial deflection in
the regular bipyramid (a non-icosahedral Johnson solid) does NOT appear on
the actual icosahedral edge path. The equatorial -1/sqrt5 belongs purely to
the TAU face mode, not the muon edge mode.
```

3. ELECTRON (VERTEX MODE)

Fully derived in doc_jobson_cell (J26); independently re-verified here as LM1 (this doc's own companion script, using m_p=938.272046 MeV) vs jobson_cell_doc.py's J26 (using m_p=938.272088 MeV) -- both are the same CODATA proton mass at different rounding/vintage, not a derived quantity (m_p remains the framework's one external input). The tiny m_p difference is why the two scripts' quoted precision differs slightly (+0.000065% here vs +0.00007% for J26 directly, Section 8) -- both are cross-checks of the same formula, not two different claims.

```
Path: (1,2) Hopf winding on the 12-vertex icosahedron.
      12 bounces per circuit, each at cos(72 deg).
eff_e = phi = (1+sqrt5)/2 = 1.618
L3_e  = (phi^3 + log5^3) / (phi^2 + log5^2)   [log5 = ln(5), natural log]
m_e   = 2*pi * alpha^2 * phi * m_p * (1 + dn_e/pi) * (1 + 3/4*alpha^2)
      = 0.510999 MeV  (+0.000065%)  [LM1]
```

The (3/4)*alpha^2 correction: from E+ x E+ = A(1) + T1(3) (T2 absent, so no shear-mode cancellation) -- coefficient = dim(T1)/dim(E+xE+) = 3/4 directly [ih_double_cg.py DC2]. This is UNIQUE to the electron.

Polygon normalization: 12*tan(pi/12) = 3.215 ~ pi (2.3% difference). The electron uses pi as an approximation; the dn_e/pi term absorbs the residual. For the muon and tau the polygon constant differs from pi by 15.6% and 0.83% respectively.

4. MUON (EDGE MODE)

4.1 G32 restricted to D5

G32 restricted to the D5 subgroup of I:

```
G32 | D5 = E1 + E2   (exact, integer multiplicities)
```

E1 and E2 are the two roots of x^2 + x - 1 = 0 (shifted golden quadratic):

```
E1: C5 char = 1/phi (sub-resonant)
E2: C5 char = -phi  (anti-resonant)
Together: C5 sum = (1/phi) + (-phi) = -1 = G32 C5 character.
```

Electron E+ is the root of x^2 - x - 1 = 0 (classical golden equation, norm = sqrt5).

Muon G32 is the Galois-pair bound state from x^2 + x - 1 = 0 (shifted by x -> -x).

4.2 Path geometry

The muon's path follows EDGES of the icosahedron in a zig-zag:

```
top_apex -> upper[0] -> lower[0] -> bottom_apex -> lower[2] -> upper[2] -> top_apex
(6 vertices, 5 bounces, all following actual icosahedral edges of length L_J)
```

Every bounce deflects by cos(72 deg) = 1/(2*phi) -- UNIFORM C5 type.

No equatorial deflection appears. All turns are identical.

The pentagon polygon normalization uses N=5 (the symmetry order of the zig-zag):

```
C_mu = 5*tan(pi/5) = 3.6327    (15.6% above pi -- cannot approximate as pi)
```

4.3 Born balance

```
eff_mu = (9-sqrt5)/8 = (1 + sqrt5/(2*phi))/2 = 0.845492
[Derived from bipyramid deflection ratio; exact icosahedral Born balance open.
NOTE: this is one of FOUR distinct candidate values in the repo, none
derived from first principles without an external input: (1) 0.845492 here
(bipyramid analogy, used in the mass formula below); (2) eff_mu=0.856308
(koide_proof.py KP1/KP2, numerical inversion against the PDG-measured muon
mass -- about 1.3% higher than (1)); (3) eff_mu=1.0 (muon_internal_force_check.py
F1-F3, recomputing the Born-ratio on this document's own corrected uniform-
72-deg path, Section 4.2 -- the ratio trivially collapses to 1 since the real
path has only one deflection type, suggesting the Born-RATIO construction
itself may not carry over cleanly from the mixed-mode bipyramid model to the
pure single-mode path); (4) eff_mu = 1/phi + alpha*(m_mu/m_e)^(2/3) = 0.873199
(muon_medium_plus_area_combined.py, additive combination of the G32|D5
sub-resonant root 1/phi, Section 4.1, with a medium-density-style correction
of order alpha scaled by the muon/electron mass ratio to the 2/3 power --
gives +0.0046%, comparable to but not better than (1); a robustness scan,
muon_combined_robustness_scan.py, shows this sits in a smooth, non-fragile
region of (exponent, coefficient) space rather than an isolated fitted
point, but the specific exponent 2/3 and coefficient (delta=alpha exactly)
were selected by testing a small number of options, not independently
derived). Which of these (if any) is the physically correct input is not
settled.]

L3_mu = (eff_mu^3 + log5^3) / (eff_mu^2 + log5^2)  = 1.4442
k_mu  = alpha * eff_mu * (1 - 3/4*alpha^2) / (1 + x_mu + x_mu^2)  = 0.006137
dn_mu = L3_mu * k_mu                                                = 0.008864
```

4.4 Mass formula

```
m_mu = 2*pi * alpha * (2/sqrt5) * phi^2 * m_p
       * (1 + dn_mu/(5*tan(pi/5)))
       * (1 + Rs^2 + 2*alpha)
     = 105.655 MeV  (PDG: 105.658 MeV,  -0.003%)

Ingredients:
  alpha^1 (not alpha^2): one Born-vertex loop (edge mode, not vertex)
  (2/sqrt5): q-component of unit (1,2) winding vector (1,2)/sqrt5
  phi^2: extra phi vs electron from G32 needing both C5 and C3 coupling
  1 + Rs^2 + 2*alpha: Rs^2 = T_2g shear mode jamming at Maxwell critical (3V-E=6)
                      2*alpha = free spin (2 transverse channels)

[lepton_mass.py LM6-LM8, PASS]
```

5. TAU (FACE MODE)

5.1 I52 path geometry -- two corpuscle photons on the Hamiltonian circuit

CORRECTED (2026-08-28): the earlier description of the tau as a single corkscrew "spiraling up through hemisphere faces" was the stale bipyramid model. The correct picture:

TWO CORPUSCLE PHOTONS (in the corpuscle model -- see doc_qm.txt FRAMING NOTE, v4 2026-08-27, for the corpuscle interpretation used throughout this section) on the same Hamiltonian circuit (the unique 20-face cycle on the icosahedral face-adjacency graph, TPC1c) traveling in opposite directions at speed c. Each photon scatters at face-CENTER nexuses -- impact-rebound at 72 deg (C5, GH2) off the convergent gluon maximum (all three gluons of that face reaching amplitude peak simultaneously, GH0b).

MEETING GEOMETRY: the two photons meet TWICE per circuit -- at hop 0 (start

nexus) and hop 10 (the face-center nexus diametrically opposite on the
circuit). Between meetings each photon traverses 10 face-center nexuses;
together they cover all 20.
The N=2 tau count itself is established independently of the flawed full
simulation below: tau_necessity_verify.py's TN1 exhaustively searches every
(starting face, initial direction) pair against the fixed local exchange-
angle rule alone (icosahedral geometry only -- no gluon model, no assumed
backward-path shortcut) and finds exactly 1 valid 20-face completion per
start, i.e. exactly 2 directed circuits (fwd + bwd) [TN1a-b].
NOTE: what the two photons actually DO at the meeting nexus -- continue,
reverse, or redirect -- remains OPEN. The broader corpuscle-billiard
simulation (cell_cycle_billiard_v2.py, and TN4/TN5's gluon-timing
arguments in the same file) is NOT currently a trusted reference for that
mechanism: it defines the backward tau's role in the wider simulation as a
reindex of the forward path rather than an independently-simulated bounce,
and is inconsistent about the gluon count per face (6 vs 3). Pending a full
force-vector resimulation (planned billiard v3), no claim about
photon-photon interaction at the meeting nexus should be treated as derived.

PATH GEOMETRY: between nexuses the photon travels as a STRAIGHT-LINE CHORD
through the cell interior. At mid-hop: r = 0.706 x L_J, which is 0.103 x L_J
inside the edge midpoints (r_mid = 0.809 x L_J). The tau never contacts the
edge -- the chord passes beneath it geometrically (alpha=0 is a consequence,
not an assumption). [doc_jobson_cell Section 7.1, TPC3/TPC4/TPC6]

The 20-bounce count (10 per photon between meetings) gives the tau its large
mass despite a lower per-bounce coupling than the muon: N=20 vs N=5 bounces.

Two angles -- SEPARATE measurements:

```
Geometric turning angle at nexus:  72 deg (C5, GH2) -- the impact-rebound angle
Born balance coupling angle:       arccos(-1/sqrt5) = 116.57 deg (backward bias,
                                   eff_tau interaction, distinct from path geometry)
Face dihedral at edge crossings:   138.19 deg (JC5) -- surface geometry of the
                                   icosahedron at each edge the chord crosses

TAU PATH GEOMETRY [gluon_tau_helix.py GH1-GH3, CONFIRMED]:
  Hamiltonian cycle on face-center graph: ALL 20 steps = 2*phi/3 (exact),
  ALL geometric deflections = 72.000 deg (EXACT) = same C5 angle as muon.
  The -1/sqrt5 (Born balance angle) is SEPARATE from the geometric path:
    Geometric path deflection = 72 deg (forward, C5 type)
    Born balance coupling angle = arccos(-1/sqrt5) = 116.57 deg (backward)
Polygon normalization: C_tau = 20*tan(pi/20) = 3.168  (0.83% above pi)

cos(theta_dihedral_bipyramid) = -1/sqrt(5)     [bipyramid equatorial, used in eff_tau]
NOTE: actual ICOSAHEDRAL face-face dihedral = arccos(-sqrt(5)/3) = 138.19 deg [JC5]
      arccos(-1/sqrt5) = 116.57 deg is the DODECAHEDRON dihedral (dual polyhedron)
```

This is the SAME -1/sqrt5 as the bipyramid's equatorial deflection -- not a
coincidence: the bipyramid equatorial crossing IS an icosahedral face dihedral.
But it belongs to the TAU (face mode), not the muon (edge mode).

5.2 Born balance

```
eff_tau = (1 + |cos_dihedral|)/2 = (1 + 1/sqrt5)/2 = 0.7236
[Estimate; exact Born balance on 20-face graph is open]

Comparison of eff values:
  eff_e   = phi              = 1.618  (vertex, positive C5 -- constructive)
  eff_mu  = (9-sqrt5)/8      = 0.845  (edge, mixed-mode from bipyramid approx)
  eff_tau = (1+1/sqrt5)/2    = 0.724  (face, negative dihedral -- inward)

The eff decreases from vertex to edge to face: the steeper (more negative) the
deflection, the lower the eff, the lower the Born coupling amplitude. Yet the
tau is the HEAVIEST lepton because it has N=20 bounces vs N=5 for the muon --
the increased bounce count more than compensates the lower coupling per bounce.
```

5.3 Mass formula (partial)

```
m_tau (Koide, from derived m_e and m_mu): 1776.92 MeV (+0.004%)

Direct formula (leading order, no corrections):
  base_tau = (1/sqrt5) * phi^3 * m_p = 1777.49 MeV  (+0.035%)  [LM16]

Correction structure: OPEN. The tau winding has no Born vertex contact
(alpha^0, no 2*pi), so corrections cannot simply copy the muon's structure
(1 + Rs^2 + 2*alpha). The Rs^2 term (T_2g shear) likely still applies since
the tau traverses T_2g faces, but the coupling mechanism differs from the
edge-mode Maxwell jamming. The exact tau correction requires the Hopf face
flux integral. Koide is the best current constraint on the corrected result.

WINDING FACTOR -- corkscrew p-component:
  Electron winding: 2*pi * alpha^2 * phi^1   (circulation × 2 Born contacts)
  Muon    winding: 2*pi * alpha^1 * (2/sqrt5) * phi^2  (circulation × 1 Born)
  Tau     winding: (1/sqrt5) * phi^3  (flux, NO 2*pi, NO alpha)
    - No 2*pi: face coupling is a FLUX through the face normal, not a closed loop
    - No alpha: tau corkscrew never contacts a vertex or edge; dihedral crossings
      have no Born vertex contact (face interior = no lattice-point coupling)
    - (1/sqrt5) = p-component of unit (1,2) winding vector (muon used q=2/sqrt5)
    - phi^3 = 2+sqrt5  (exact algebraic identity)

base_tau = (1/sqrt5) * phi^3 * m_p = (phi^3/sqrt5) * m_p
         = 1777.49 MeV  (+0.035% from PDG 1776.86 MeV)  [LM16]

ALGEBRAIC IDENTITY (exact):  phi^3/sqrt5 = 1 + 2/sqrt5 = 1 + (muon winding)  [LM17]
  The tau winding factor is exactly 1 (face floor/ceiling contact) plus the muon's
  longitudinal winding factor (2/sqrt5). The tau corkscrew CONTAINS the muon spiral.

The 0.035% residual closes when the exact Hopf fiber flux integral through one
icosahedral face is computed. Koide constrains the corrected result to +0.004%.

ELASTIC MEDIUM INTERPRETATION:
  m_tau = (phi^3/sqrt5) * m_p encodes the tau's COUPLING TO the T_2g elastic
  face stiffness. The T_2g shear field creates the face material (color-neutral,
  elastic); the tau corkscrew rides it. The tau mass = energy cost of the
  corkscrew mode propagating through the T_2g face surfaces.
  The Rs^2 correction in the muon formula = T_2g shear amplitude at Maxwell
  critical (3V-E=6); the same T_2g field that the tau directly traverses.

Near-critical note: N_J(tau) = hbar_c/m_tau/L_J ≈ 11 (vs Maxwell critical 21).
The tau is closer to the Maxwell critical than the muon (N_J~188). This may add
a near-critical correction not present in the muon formula.

[lepton_mass.py LM10, Koide result PASS]
```

6. KOIDE FORMULA

6.1 Numerical result

```
(m_e + m_mu + m_tau) / (sqrt(m_e) + sqrt(m_mu) + sqrt(m_tau))^2 = 2/3

With PDG masses: 0.66666051  (2/3 = 0.66666667,  deviation = -9.2 ppm)
With derived m_e (+0.000065%) and m_mu (-0.003%):
  m_tau (Koide) = 1776.92 MeV  (PDG 1776.86 MeV,  +0.004%)
  Koide ratio (derived masses) = 0.66666667  (exact 2/3 to 10 decimal places)

The tau prediction inherits the muon's -0.003% error through the Koide formula.
This is a TAUTOLOGICAL check, not an independent confirmation: m_tau here is
obtained by algebraically SOLVING the Koide equation for m_tau given the
derived m_e/m_mu (LM10), so verifying that result satisfies Koide (LM11) is
guaranteed by construction. It does not, by itself, show the muon/tau
residuals "close" under an independently exact Born balance -- that remains
the open item in Section 9.
```

6.2 Why Koide should be exact -- geometric origin and proof requirements

GEOMETRIC ORIGIN (new -- face_gluon_geometry.py FG11):

```
The Koide 2/3 is the FIELD/FIELD-STRENGTH RATIO of the icosahedral cell:

  dim(T_1g) + dim(T_2g)     3 + 3     6
  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ = ■■■■■■ = ■■■■ = 2/3  [EXACT]
     dim(T_1g x T_2g)         9       9

T_1g (transverse, W/Z field):     dim = 3
T_2g (shear, elastic face field): dim = 3
T_1g x T_2g = G + H (field strength F_mn, H_g = gluon flux lines): dim = 9

The icosahedral group I_h has two 3-dimensional irreps (T_1g and T_2g)
whose product has dimension 3*3 = 9, giving ratio 6/9 = 2/3. NOTE: this
dimension count alone is NOT I_h-specific -- dim(V⊗W)=dim(V)*dim(W) holds
for ANY two 3-dim irreps of ANY group (e.g. the octahedral group O's own
T1/T2 irreps give the identical dim-9 product; no script in this repo
checks uniqueness against other groups, and a by-hand check for this audit
found O is not excluded by this count alone). What IS icosahedron-specific
here is T_1g x T_2g decomposing with NO A_g component (G+H only, verified
by projection formula). The three leptons satisfy Koide because they are
Born-balanced modes of this same icosahedral cell -- but the connection
from this dimension-ratio fact to why the derived lepton MASSES sum to
exactly 2/3 is the separate, still-open step described next.

Additionally: G32 x G (muon x gluon) = G32 + 2*I52, with NO A component.
The muon is color-neutral (cannot emit/absorb a single gluon), consistent
with leptons being outside the color sector. The muon rides gluon edge
channels geometrically via C3=+1 coupling, not via direct vertex coupling.
```

NUMERICAL STATUS (koide_proof.py, 8/8 PASS):

```
The leading-order mass formulas give K0 = 0.67111 (4442 ppm(abs) above 2/3 --
an absolute shift in K, not the normalized relative-ppm convention used
elsewhere in this paper for final precision claims, e.g. the 9.2 ppm(rel)
Koide agreement below).
The Born corrections must decrease K by this amount to reach 2/3.
This large shift is now understood: the corrections enforce the cell's
geometric 2/3 ratio (field/field-strength) on the lepton Born balances.
```

WHAT THE FORMAL PROOF STILL REQUIRES:

```
(1) Exact Born balance eff_mu for the 5-bounce uniform-72-deg icosahedral path
(2) Hopf fiber flux integral over one icosahedral face (exact base_tau)
(3) Show the three exact mass expressions satisfy 4(ab+bc+ca) = a^2+b^2+c^2
The geometric origin (FG11) is proven; the algebraic connection to the Born
balance correction terms is the remaining open step.
```

7. COLOR COUPLING AND F-7

7.1 I52 dim=6 = 2 x 3

The icosahedron's 20 faces are 3-colorable: assign colors R, G, B such that no two adjacent faces share a color. This 3-coloring is identified with quark color charge (open item F-7). The tau (I52) traverses ALL 20 faces, coupling to all 3 colors with equal GROUP-THEORETIC weight (C3=0) -- though the physical face count itself is 7R/7G/6B, not exactly equal (20 is not divisible by 3; su3_from_faces.py SU3-8; face_gluon_geometry.py's own comment: "cannot be perfectly divided into 3 equal color groups"). Since C3=0 in I52, it couples to R, G, B with equal amplitude
and zero net color charge:

```
I52 dim = 6 = 2 (Dirac spinor) x 3 (quark color from face 3-coloring)
```

7.2 The three C3 characters and color roles

```
E+  C3 char = -1  (vertex): 5 colored faces meet at each vertex.
                            Anti-phase C3 -> electron is a color SINGLET
                            (destructive interference of all 3 colors at vertex)
G32 C3 char = +1  (edge):   2 colored faces share each edge.
                            In-phase C3 -> muon couples to face BOUNDARY
                            (net color at the interface between two colored regions)
I52 C3 char =  0  (face):   1 colored face = pure color-neutral by C3.
                            Zero C3 -> tau couples to full color content of each face
                            but the full cycle through all 3 colors averages to zero
```

The muon's C3=+1 is the geometric origin of its sensitivity to the quark-gluon boundary structure (consistent with mu+ capture in nuclei and muon-catalyzed fusion coupling). The electron's C3=-1 singlet character is why it does not couple to color.

8. COMPARISON TABLE

Property	Electron (E+)	Muon (G32)	Tau (I52)
2I irrep (dim)	E+ (dim=2, j=1/2)	G32 (dim=4, j=3/2)	I52 (dim=6, j=5/2)
C3 character	-1	+1	0
Icosahedral mode	VERTEX	EDGE	FACE
Element count	V = 12	E = 30	F = 20
Deflection	cos(72 deg) = 1/(2*phi)	cos(72 deg) = 1/(2*phi)	GEOMETRIC: 72 deg (C5)
(Born balance)	eff from cos(72)	eff from cos(72)	eff from cos(-1/sqrt5)
Deflection bias	forward (cos > 0)	forward (cos > 0)	path: forward (72 deg)
Bounce count N	12	5	20
Polygon N*tan	3.215 (~pi)	3.633 (not pi)	3.168 (~pi)
eff parameter	phi = 1.618	(9-sqrt5)/8 = 0.845	(1+1/sqrt5)/2 = 0.724
Alpha power	alpha^2	alpha^1	alpha^0 (no Born contact)
phi power	phi^1	phi^2	phi^3
Free-spin corr	3/4 * alpha^2	Rs^2 + 2*alpha	Rs^2 + 2*alpha (+ near-crit?)
T2 cancellation	no (T2 absent)	yes (T1=T2 in G32xG32)	yes (T1=T2 in I52xI52)
Color role	SINGLET (intersection)	BOUNDARY (edge)	NEUTRAL (face=T_2g surface)
Mass derived	0.510999 MeV (+0.00007%)	105.655 MeV (-0.003%)	1776.92 MeV (+0.004%)
PDG value	0.510999 MeV	105.658 MeV	1776.86 MeV
Path formula	J26 in doc_jobson_cell	Section 4 above	corkscrew: phi^3/sqrt5 * m_p

The bipyramid model bridges muon and tau:

```
Bipyramid apex (72 deg)      = muon deflection  -> pure muon path uses ONLY these
Bipyramid equatorial (116.57 deg) = tau deflection -> pure tau path uses ONLY these
Mixed bipyramid eff = (9-sqrt5)/8 = Born balance BETWEEN the two modes
```

9. STATUS

DERIVED (zero free parameters, all verified by companion scripts):

- All three lepton = 2I spinors, assigned by spinor rank [exact, group theory]
- Electron = VERTEX, muon = EDGE, tau = FACE (icosahedral element mapping) [LM12-LM13]
- Bipyramid was superposition of muon (apex) and tau (equatorial) modes [Section 2.3]
- m_e = 0.510999 MeV (+0.000065%) J26 formula [LM1]
- m_mu = 105.655 MeV (-0.003%) pentagon Born balance, using the bipyramid-approximation eff_mu -- known to be ~1.3% from the numerically-inverted value, and the Born-ratio construction itself is not yet settled (three unreconciled candidates; see Section 4.3 caveat) [LM6-LM8]
- m_tau = 1776.92 MeV (+0.004%) Koide from derived m_e, m_mu [LM10]
- Tau has a nexus with the gluon at each face boundary (20 per circuit) [LM14]
  Born balance deflection = arccos(-1/sqrt5); actual face-face dihedral 138.19 deg [JC5]
- I52 dim=6 = 2 x 3 = spinor x quark colors [LM15]
- Tau corkscrew base = (1/sqrt5)*phi^3*m_p = 1777.49 MeV (+0.035%) [LM16]
- phi^3/sqrt5 = 1 + 2/sqrt5 (exact algebraic identity) [LM17]
- Tau winding: alpha^0 (no Born vertex contact), no 2*pi (face flux not loop)
- T_2g = color-neutral elastic face material; Rs^2 = T_2g shear amplitude [FG4,FG5]
- 2G (8-dim, C3=+1) = 8 gluons from Gamma(20 face centers) decomp [FG3]
- Tau rides T_2g surface; does not constitute it [FG7]
- H_g = T_1g x T_2g (field strength F_mn, not a new particle) [FG8]
- Muon rides gluon edge channels; 72-deg = gluon flux bending at C5 vertex [FG9]
- G32 x G = G32 + 2*I52 (no A): muon is color-neutral in gluon sector [DG15]
- KOIDE 2/3 geometric origin: dim(T1g+T2g)/dim(T1gxT2g) = 6/9 = 2/3 [FG11]
- F-7: SU(3) from icosahedral face 3-coloring [CLOSED 2026-08-23, su3_from_faces.py 11/11]

ESSENTIALLY CLOSED:

- Koide 9.2 ppm from measured masses [LM9, independent -- uses PDG masses directly]; Koide EXACTLY 2/3 from derived masses [LM10-LM11] -- this second check is TAUTOLOGICAL, not independent: m_tau here is obtained by algebraically SOLVING the Koide equation for m_tau given derived m_e/m_mu, so it satisfying Koide is guaranteed by construction, not a confirmation comparable in kind to LM9's PDG-mass agreement.
- No alpha^2 correction for muon or tau (T1=T2 in G32xG32, I52xI52) [Section 1.2]
- Color coupling from C3 characters (E+=-1, G32=+1, I52=0) [Section 7]
- Euler V-E+F=2 and Maxwell 3V-E=6 constrain all three path geometries [LM12-LM13]
- Lepton generations exhaust 2I spinors (3 independent spinor irreps)
- KOIDE GEOMETRIC ORIGIN: 2/3 = (dim T1g+T2g)/dim(T1g x T2g) = 6/9 [FG11, proven]
- Muon color-neutrality: G32 x G has no A component (color-neutral by group theory) [DG15]

OPEN:

- Exact muon Born balance for uniform-72-deg icosahedral path (eff_mu=0.8563 numerical)
- Exact Hopf fiber flux integral through one icosahedral face (closes tau +0.035%)
- Tau correction structure (no Born vertex contact -- differs from muon's Rs^2+2*alpha)
- Born balance algebraic closure: show 4442 ppm correction enforces 6/9 ratio [KP7]
- Near-critical tau correction (N_J~11, Maxwell critical ~21)
- Tau meeting-nexus mechanism at hop 0/10: what the two photons actually do when they coincide (continue, reverse, redirect) is OPEN. (The N=2 tau count itself is NOT part of this open item -- it is independently established by TN1's exchange-angle-rule enumeration, Section 5.1.) The broader corpuscle-billiard simulation is NOT a trusted reference for the meeting mechanism (undisclosed tau-backward shortcut in the wider sim, inconsistent gluon-per-face count); needs a full force-vector resimulation (planned billiard v3) before any claim here is derived.

WHY THREE GENERATIONS (no more, no fewer):

The 2I double group has exactly 4 spinor irreps: E+(2), E-(2), G32(4), I52(6). Treating E+/E- as particle/antiparticle, there are exactly 3 independent free fermionic modes. These ARE the 3 lepton generations. No 4th exists in 2I.

10. COMPANION SCRIPTS

```
analysis/demos/leptons_doc.py         -- 18/18 PASS  [STANDALONE -- no project imports; uses numpy for vector geom
analysis/quantum/lepton_mass.py       -- 18/18 PASS  (main: all three leptons, exact formulas)
analysis/quantum/face_gluon_geometry.py -- 14/14 PASS (T_2g, 2G, H_g, Koide origin, lattice geometry)
analysis/quantum/koide_proof.py            -- 8/8 PASS  (Koide gap, eff_mu inversion, geometric origin 2/3=6/9 KP1
analysis/nuclear/ih_double_group.py   -- 15/15 PASS (2I character table, spinor irreps, G32xG=G32+2*I52 DG15)
analysis/quantum/gluon_tau_helix.py    -- 10/10 PASS (tau path geometry, GH0-GH0c, GH6-GH7)
analysis/quantum/tau_pair_configuration.py -- 11/11 PASS (TPC1-TPC6: tau circuit uniqueness, two-photon
                                             meeting geometry -- NOT currently trusted, see Section 9 OPEN item)
```

```
analysis/corpuscle/tau_necessity_verify.py -- 12/12 PASS (TN1 only: N=2 tau count independently
                                              established by exchange-angle-rule enumeration, trusted;
                                              TN4/TN5 gluon-timing claims NOT trusted, see Section 9 OPEN item)
analysis/quantum/su3_from_faces.py     -- 11/11 PASS (F-7: SU(3) from face 3-coloring)

LM1-2:   Electron J26 reference (+0.000065%)
LM3-5:   Pentagonal bipyramid deflection geometry (exact, h_t/r_e = 1/phi)
LM4b:    Actual icosahedral path: all deflections = 72 deg (uniform C5)
LM6-8:   Muon mass from pentagon Born balance (-0.003%)
LM9:     Koide formula (9.2 ppm)
LM10-11: Tau from Koide (derived m_e, m_mu, +0.004%)
LM12-13: Icosahedral Euler/Maxwell constraints (V=12, E=30, F=20)
LM14:    Tau dihedral = arccos(-1/sqrt5) (exact)
LM15:    I52 dim=6 = 2*3 (spinor x quark colors)
LM16:    Tau corkscrew base = phi^3/sqrt5 * m_p (+0.035%, leading order; correction open)
LM17:    phi^3/sqrt5 = 1 + 2/sqrt5 (tau contains muon winding, exact)

FG1-3:   Gamma(20 face centers) = A+T1+T2+2G+H; 2G=8 gluons
FG4-5:   T_2g C5=-1/phi (shear), Rs^2 = T_2g shear amplitude at Maxwell critical
FG6-7:   G C3=+1 (gluon color-active), T2 C3=0 (face elastic, color-neutral)
FG8:     T1g x T2g = G + H (H_g = field strength, not a new particle)
FG9:     Muon 72-deg deflection = gluon edge-channel bending at vertex (exact)
FG10:    G32 (muon) and G (gluon) share C3=+1 -- C3 coupling locks muon to gluon channels
FG11:    Koide 2/3 = (dim T1g + dim T2g) / dim(T1g x T2g) = 6/9  [EXACT]
FG12-14: V=12 from spinor sum 2+4+6; E=30 from C5; F=20 from Euler V-E+F=2

DG15:    G32 x G = G32 + 2*I52, no A (muon color-neutral; ih_double_group.py 15/15 PASS)

KP1-2:   Exact eff_mu = 0.8563 by inversion (bipyramid approx 0.8455 = 1.3% off)
KP3:     base_tau/m_p scenario B vs phi^3/sqrt5 (0.28% off: needs Hopf flux)
KP4-5:   Koide holds to 9.2 ppm; algebraic form a^2+b^2+c^2=4(ab+bc+ca) to 0.003%
KP6:     Koide texture angle theta = 132.73 deg; m_e reconstruction to 0.001%
KP7:     Leading-order K0 = 4442 ppm above 2/3 (proof needs exact Born balances)
KP8:     Geometric origin: 2/3 = (dim T1g+T2g)/dim(T1g x T2g) = 6/9 exactly [FG11]
```

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[doc_nucleus]      https://doi.org/10.5281/zenodo.22042337  [2I CG, irrep assignments]
[doc_leptons]     https://doi.org/10.5281/zenodo.22057421  [this document]
[doc_muon.txt]     superseded by this document
[doc_tau.txt]      superseded by this document
```


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Repository: https://doi.org/10.5281/zenodo.22105622

Code: https://github.com/Destraag/torsionverse_public